

ABRA HANDOFF — 2026-08-04, the session R4 decided

Paste this into a new session. Written to be read by someone with no memory of the night. Supersedes nothing: `docs/HANDOFF-2026-08-04.md` is the handoff this session *received*.

o. THE ONE THING THAT CHANGED

R4 DECIDED. MILTANK is measurably stronger than MAG.

```
535 decisive pairs · MILTANK takes 55.5% · log-likelihood 3.00 vs a 2.94 bound
==> Accept H1 at alpha=0.05.          data/games.r4-decided.jsonl (5,248 games)
```

The pre-fix baseline on the **broken** engine was **-0.28** — drifting toward *no better*. Same search, same flags, same everything. The only change was the model.

That is the result to build on: a search is worth exactly what its model is worth. Closing MEDICHAM's gaps now has a measured return rather than a hoped-for one.

Do NOT quote R4 as "the bot is good". It says the SEARCH beats the fitted policy, at `--miltank-n 30`, with uniform-random playout opponents, and with the preview search disabled (sheets arrive after preview inside `mew`). It is a floor, not a description.

1. HARD RULES — from Will, not negotiable

- **Never** `git add -A` **or** `git add -u`. The ingest churns `archetypes.json`, `kad-replays.js`, `live.js`, `conformance.json`. Add by name. Violated once this session and backed out.
 - `SHOWDOWN_PATH=C:/Users/willj/Projects/Pokemon/pokemon-showdown` is required for almost everything.
 - `fit_policy.js` / `fit_joint.js` need `node --max-old-space-size=4096`.
 - **6 processes max**. Ask before running wide while he is at the keyboard.
 - **Never edit** `board.js`, `magnemite.js` **or** `engine-data.js` **while a fit or self-play run is in flight**. And **never restart the live bot mid-battle** — it forfeits his game. I did it once.
 - **A REFIT INVALIDATES SEVEN ARTIFACTS**: counterplay, winrate-backtest, opponent-calibration, weight-multiplicity, then the mag/mew/scoreboard bundles. `provenance.js` now DERIVES this — it tracks engine code as an artifact input, so those seven flag themselves.
 - A restamp is only valid when the feature FUNCTION is unchanged. If `board.js` or the damage table changed, **REFIT**.
 - **MEASURE THE NOISE FLOOR** before believing any effect. **CHECK THE CORPUS STAMPS** before attributing an effect to a lever.
 - **SPRT-gate every H2H and read it AS IT GOES**. Levers are PER ARM. Arm 1 is the challenger.
 - **No hardcodes** — derive from handlers and artifacts. A boolean is not a parameter.
 - OTS only: games without open team sheets are thrown out; OTS games are recorded and the replay auto-published to `data/live-games/replays.txt`.
 - Password lives in `data/.showdown-pass` (gitignored). **Never type it into a command**.
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2. RUNNING RIGHT NOW

MAGABRA	wss://sim3.psim.us , --miltank --rollout-n 120 --meta-teams 0.08 --trash --trace --port 8111
A/B in flight	6 shards, --miltank-foe prior vs uniform → data/.mew-shards/foe-s*.jsonl
Head-to-head vs Will	~15–4 to Will. He is deliberately playing unfamiliar teams — do not read it as strength.

Read a run: `node engine/sprt.js <file>` . Combine shards by `cat` -ing them first.

MAGABRA is LOCKED on Showdown (`!magabra` in PMs). It can battle and save replays but **cannot chat**, so `--trash` is wired and silently dropped. Not a code bug — a flagged-IP account lock.

3. THE CAST

Name	What it is
MEDICHAM	<code>medicham2-browser.js</code> — fast approximate battle engine, ~1.7 ms a full game. Everything MILTANK believes comes from here.
MAG	<code>magnemite.js</code> — fitted linear policy over 58 <code>board.js</code> features, trained to predict what a HUMAN clicks. Imitates.
MILTANK	<code>miltank.js</code> — the SEARCH player. Plays positions out, takes the line that wins most. Named by Will after the classic Rollout user. See <code>docs/MILTANK.md</code> .
<code>board.js</code>	a position → 58 numbers. MAG's eyes. Computes damage THROUGH MEDICHAM, so damage bugs reach MAG and force a refit.
MEW	<code>mew.js</code> — self-play factory and the only harness that can run a controlled A/B.
abra-tags	Will's rulebook: 172 tags over every move/ability/item. The knowledge is complete; the implementation is not.

MILTANK makes three decisions: **bring/lead** (90 brings played out against the opponent's sheet), **both clicks each turn** (successive halving), **post-KO replacement**. It defers to MAG only when the finalists sit inside one standard error — a wider band near the ceiling.

4. INSTRUMENTS BUILT THIS SESSION

File	Question
<code>tests/test-mechanics.js</code>	does the engine actually DO this? 54 probed, 42 live, 12 missing
<code>tests/walk_tags.js</code>	every tag vs Showdown — the tag says WHAT, Showdown says what SHOULD happen
<code>tests/test-engine-diff.js</code>	MEDICHAM vs Showdown on real matchups; finds bugs nobody imagined
<code>tests/mechanics_rank.js</code>	which unread tag costs the most, by corpus usage
<code>tests/mechanics_surface.js</code>	upper bound on what is unwired — weak, over-reports, and says so

`data/mechanics-census.json` holds the count. `live` **must never go down.**

5. STILL OPEN

Confirmed missing (census): Lightning Rod · Tri Attack · White Herb · Cursed Body · Facade · Feint through Protect · Giga Impact recharge · Avalanche · Gyro Ball · Belly Drum · Ice Scales · Unnerve

Found by the differential tester, not yet in the census: Freeze-Dry (deals *less* than Ice Beam into Water — the move's whole identity) · Haze · Friend Guard · Poison Touch · Gigaton Hammer · Expanding Force · Marvel Scale · Disguise

Structural:

- **Leaf calibration** — reads 100% and loses, reads 1% and wins. Every MILTANK decision is an argmax over these numbers. `backtest_winrate.js` asks exactly this and has never been pointed at the current leaf. **Biggest remaining bug.**
- **Opponent model** — playouts use uniform-random moves. Real Charizard clicks Protect 60.6%, not 25%. `--miltank-foe prior` exists; the A/B is running.
- **Which mega to take** — currently "the lead keeps it", an arbitrary tiebreak. Should be a search decision; only two-stone brings would branch. "Biggest stat gain" was measured and DISCARDED (every Champions mega is +101 to +104).
- **Team quality** — `--meta-teams` gives 169 teams, but the base filter is COMPLETENESS not quality.
- **118 of 172 tags unprobed.**

6. LESSONS THAT COST HOURS

1. **A silent default looks exactly like a working feature.** The lead search never ran in a real game across FOUR layers of bug, each found only because the fallback announced itself. The post-KO search logged every turn and never once decided. **Make every fallback loud.**
2. **Search amplifies model error.** A maximiser seeks the lines its model is most optimistic about — which are the ones it is most wrong about. This is why R4 was negative on the broken engine.
3. **Usage counts are SHEET counts.** Blaze reads 4,585 and is nearly worthless — 30 of 54 entries are a Charizard that megas into Drought turn one. Ice Scales, Filter, Aerilate, Prism Armor, Punk Rock and Ripen all have **0 uses**; a stretch of work went into them before Will said so.
4. **Every derivation over-matches on the first try.** `refusesStatusMoves` caught Telepathy and Wonder Guard; `speedOnItemLoss` caught Sticky Hold; `failsIfTargetNotAttacking` caught Quick Guard, Wide Guard and Round. **Print what a new tag matched before wiring it.**
5. **My own probes were wrong ~15 times, always toward a comfortable answer.** A Corviknight ally immune to the Earthquake being blocked; a Garchomp that died before it could freeze; a probe that applied the status itself then asserted it; a "plain" Basculegion already holding a Choice Scarf. **Clear the control explicitly. Test the outcome, not the classification. Identical results across a varied knob mean the knob is unwired.**
6. **Will's domain knowledge beat the data repeatedly** — Blaze/mega, Gardevoir-is-Trace, Contrary-Staraptor, "Meganium needs the mega", "reality is not a 1/4 split for all moves". When he says a number looks wrong, **check it before defending it.**
7. **Never read an interim SPRT.** 66.7% became 44%; 57.7% became 50%. The bound exists for this.
8. **Reuse the one canonical path.** Hand-rolling a second body builder produced `buildMon("Scizor") → null`; `dmgMon` already existed and does it right.

7. WHAT I WOULD DO NEXT

1. **Read the opponent-model A/B** (running). If `prior` wins it changes every evaluation.
2. **Calibrate the leaf** — take positions it calls 90–100% and measure how many are actually won.
3. **Keep walking the tag list.** Now justified by R4 rather than hoped.

4. **Make the mega choice a search decision.**

5. **Ship an opponent-aware payout** if the A/B says so — the road to an equilibrium player.

`docs/MILTANK.md` §3.1 explains why the current best-response is exploitable by construction.

8. GLOSSARY

Mickey Mouse team — Will's term for a corpus team that is real, open-sheet and still terrible (the naming case led Pikachu AND Raichu). `--meta-teams` filters by average member usage; the pool is announced on every start, on or off.

MILTANK / MAG — search vs imitation. See §3.

R1–R4 — the rollout gates. R1 leaf accuracy (pass), R2 cost (pass), R3 divergence (pass, 72.9%), **R4 does it actually win (PASS, 55.5%)**.